HEALTHIER, LONGER, BETTER LIVES

Can OCR Rules Replace Azure OpenAI?

Medical receipt extraction benchmark

9 receipts · 6 fields · 3 approaches · plus a diagnosis-comment test

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The Answer in Brief

Rules can replace AOAI only for simple, clearly labelled tasks.

Not as a general replacement.

Rules can match AI on simple, labelled checks, after manual work.

AOAI is needed when receipt layouts vary and when richer medical information is wanted.

88.2% vs 37.3–41.2%	Before any tuning, AOAI was far ahead. The first rules had no Chinese labels.
96.1% vs 86.3–88.2%	After tuning on the same receipts (development results). Rules needed 69 → 225 label phrases; AOAI one prompt revision.
13/13 for all	On a yes/no diagnosis-comment check with clear labels, rules matched AOAI.

Development results are not independent validation; a test on 30+ unseen real receipts is still needed.

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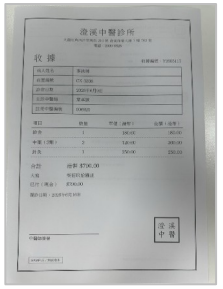
What We Compared

Three ways to turn a receipt photo into six fields. Only the extraction step differs between rules and AI.

	1. Read the text (OCR)		2. Find the fields		
PaddleOCR Light + rules	**PaddleOCR Light** open source, own compute	→	**Hand-written rules**	→	**6 fields**
Azure DI + rules	**Azure Document Intelligence** cloud OCR	→	**Hand-written rules**	→	**6 fields**
Azure DI + AOAI	**Azure Document Intelligence** cloud OCR	→	**Azure OpenAI prompt**	→	**6 fields**


English


Bilingual


Chinese

9 receipts: 7 printed test receipts (English, bilingual, Chinese layouts) and 2 from the first benchmark round.

51 fields scored: doctor, registration no., patient, receipt no., service date, total.

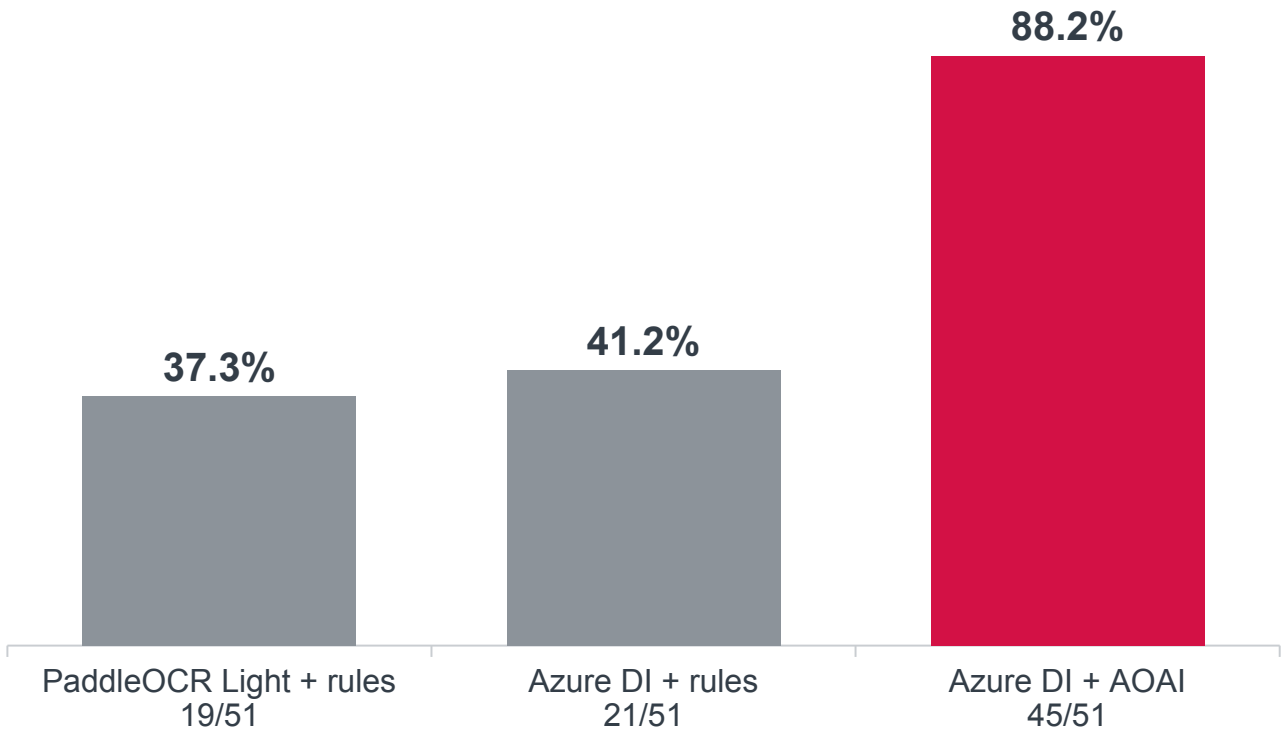
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Before Tuning: AOAI Was Far Ahead

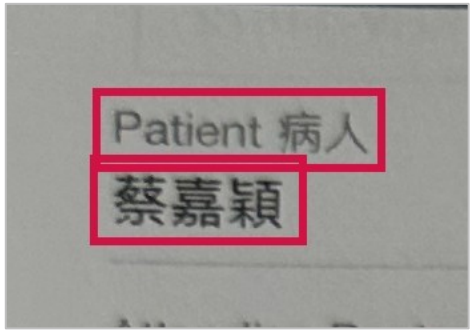
MEASURED · BEFORE TUNING

Out of the box, AI extraction found more than twice as many fields as the first rule sets.



Why the rules scored low

- The first rules knew English labels only: no Chinese or bilingual labels.
- **So this compares first versions, not the best each approach can do.**



The OCR read the name 蔡嘉穎 .

The first rules returned the label 病人 ("patient") instead.

Accuracy = correct fields out of 51 on 9 receipts. Receipt shown: bilingual medical centre receipt, PaddleOCR Light text.

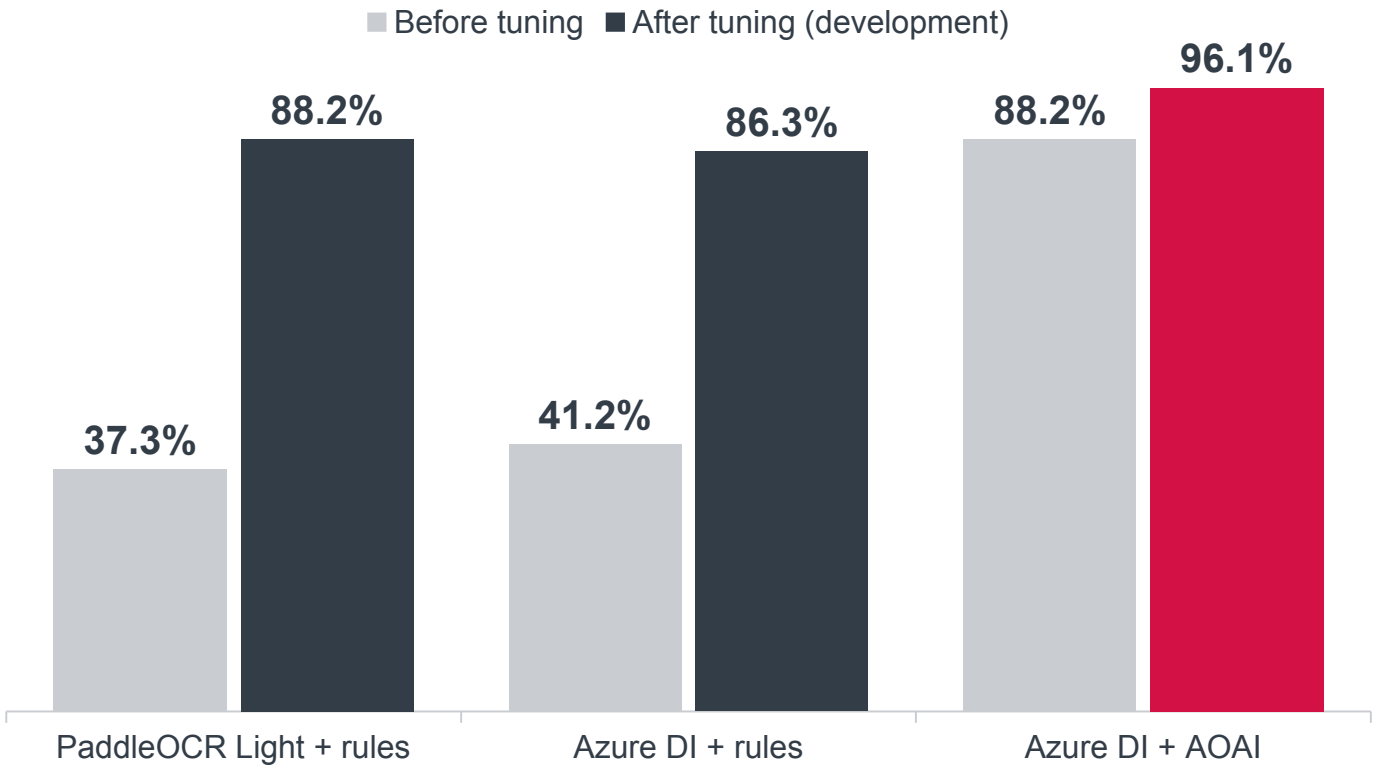
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After Tuning: All Improved, at Very Different Cost

MEASURED · DEVELOPMENT SET

Rules closed most of the gap, but only by being rewritten by hand for these receipts.



Rules: rewritten by hand

- 69 → 225 label phrases: Chinese, bilingual, layouts
- Code changes, then a retest of every receipt
- Fixed 27 fields, broke 1 that was correct

AOAI: one prompt revision

- Clearer definition of each field
- Plain-language edits, then a rerun
- Fixed 4 fields, broke 0

Development results: changes made after reviewing these 9 receipts; not independent validation. Rule figures shown for PaddleOCR Light.

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Where Errors Remain: Accuracy by Field

MEASURED · DEVELOPMENT SET

All three miss the same two fields; the rules also miss patients and receipt numbers.

Field	Receipts	PaddleOCR Light + rules	Azure DI + rules	Azure DI + AOAI, prompt v2
Doctor	9	88.9% (8/9)	88.9% (8/9)	88.9% (8/9)
Registration no.	6*	**100.0%** (6/6)	**100.0%** (6/6)	**100.0%** (6/6)
Patient	9	77.8% (7/9)	77.8% (7/9)	**100.0%** (9/9)
Receipt no.	9	77.8% (7/9)	77.8% (7/9)	**100.0%** (9/9)
Service date	9	88.9% (8/9)	88.9% (8/9)	88.9% (8/9)
Total	9	**100.0%** (9/9)	88.9% (8/9)	**100.0%** (9/9)

*Development results after tuning. * Registration number is printed on 6 of the 9 receipts; every other field on all 9.*

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What Prompt V2 Changed

MEASURED · DEVELOPMENT SET

Four fields fixed, all by clearer wording. Code checks, layout and photos added nothing.

Field fixed	Original prompt replied	Prompt v2 wording added	Prompt v2 replied
Registration no. GP receipt	✗ MBBS (HK) · Reg. No. M17642	*"the identifier only, without its label"*	✓ M17642
Doctor Dental receipt	✗ Dr. Ho Ching Yee, BDS (HK)	*"Leave out qualifications and post-nominals"*	✓ Dr. Ho Ching Yee
Doctor Pharmacy receipt	✗ Dr Jessica Sterenborg Picton Medical Centre	*"clinic, hospital or practice names"*	✓ Dr Jessica Sterenborg
Total Vet invoice	✗ HKD 1361.50	*"after any discounts, rounding or other adjustments"*	✓ HKD 1361.00

Prompt wording
4 fields fixed

Python checks
changed 0 values

OCR row layout
plain text also 49/51

Photo attached
also 49/51

Development result: prompt v2 was written after reviewing these errors. Still wrong: the vet invoice doctor and the pharmacy service date.

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Rules vs. AOAI: Side by Side

AOAI wins on accuracy and effort; rules win on cost and control. New layouts are untested.

	OCR + rules	Azure DI + AOAI	Evidence
Accuracy*	37.3–41.2% first; 86.3–88.2% after tuning	88.2% first; 96.1% after prompt v2	Measured
Speed per receipt	13.9 s PaddleOCR (CPU) · 3.8 s Azure DI	6.2 s	Measured
Running cost	PaddleOCR: own compute, no API fee · Azure DI: page fee	Azure DI page fee + ~602 input / 263 output tokens	*Prices to confirm*
Development effort	Rules rewritten by hand (69 → 225 label phrases)	One prompt revision	Measured
Side effects of changes	1 correct field broken by the rule changes	0 broken by the prompt revision	Measured
New receipt layouts	New labels and rules for each layout or language	Expected to adapt without code changes	*Expected*
Control and audit	Deterministic; every decision traceable	Depends on model and prompt; needs monitoring	*Expected*

** First = before tuning (the first rules had no Chinese labels); after tuning = development results. Green = stronger side. Expected = not measured.*

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When Rules Are Enough: Diagnosis Check

MEASURED · DEVELOPMENT SET

For a simple yes/no question with clear labels, rules matched AOAI on every receipt.

The question

Does the receipt contain a written comment about the patient's condition, symptoms or diagnosis?

13 receipts

- 3 with a comment
- 2 with an empty label
- English and Chinese labels

13/13
PaddleOCR Light + rules

13/13
Azure DI + rules

13/13
Azure DI + AOAI

What it tells us

- Rules are enough when the answer sits under a clear label (Diagnosis:, Notes:, 診斷) and the question is yes/no.
- AOAI also spotted a condition written as a line item ("broken ribs."): unlabelled detail rules cannot see.
- Adding the check to AOAI cost about 26% more input tokens and changed none of the six fields.
- But the test was easy for both: only 3 receipts had a comment, all under a label.

Development result: rules and prompt written after reading these receipts. Only 3 receipts had a comment.

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Beyond Six Fields: What AOAI Could Add

POTENTIAL · NOT YET TESTED

Our receipts already carry medical details that rules ignore. AOAI could turn them into data.

Already in our receipts' OCR text

Diagnoses and symptoms
Colitis · diarrhoea
vet invoice

Treatments and tests
Chest X-ray · Blood test (CBC) · 針灸 (acupuncture) · Scaling and polishing
hospital, Chinese medicine, dental

Medications
Omeprazole 20mg Capsules · Metronidazole 50mg Tablet
pharmacy, vet invoice

AOAI could return

- Diagnosis
- Symptoms
- Treatment or procedure
- Medication and dose
- Cost per item

Which could enable

- Claims analytics by diagnosis and treatment
- Cost benchmarks per treatment and provider
- Medication patterns
- Consistency checks: treatment vs diagnosis vs amount
- Faster triage for reviewers

Not yet tested. We have not built or measured extraction of these fields. A pilot needs ground truth, accuracy targets, a privacy review and cost checks.

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Limits of What We Know

The evidence points one way, but it is not yet proof.

1

Small sample

9 receipts, 51 fields. 7 are printed test receipts.

2

Rules started without Chinese

The first gap (37.3–41.2% vs 88.2%) overstates what rules can do.

3

Tuned on the same receipts

Later scores (rules 86.3–88.2%, AOAI 96.1%) are development results.

4

Little unseen testing

Only 4 unseen handwritten receipts: too few to compare. On all 4, Azure DI misread the year 2026 as 2020.

5

Costs not confirmed

Token counts are measured; Document Intelligence and gpt-5-mini prices are not.

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Recommendation

Not a general replacement: rules for simple, labelled tasks; AOAI where receipts vary or richer data is needed.

Simple OCR rules are sufficient when…

- Layouts are few, known and stable
- Fields have clear printed labels
- The question is simple, e.g. is a diagnosis comment present?
- No API fee or fully deterministic output is required

AOAI adds real value when…

- Many providers, layouts and languages (Chinese, bilingual)
- Fields lack consistent labels: names with titles, totals vs subtotals
- New fields should be added by changing a prompt, not code
- Richer medical data is wanted (potential, not yet tested)

1. Freeze

the improved rules and AOAI prompt v2

2. Test unseen

30+ real, anonymised receipts; count rule changes

3. Pilot extra fields

diagnoses, treatments, medications with ground truth

4. Confirm cost

Document Intelligence pages and gpt-5-mini tokens

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